
Vision-Language Models in Radiology:

Improving Trust with Uncertainty Estimation and Fact-Based Report Generation

Shayan Salehi, Ali Salesi, Sina Daneshgar, Sina Beyrami

Department of Computer Engineering
Sharif University of Technology

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Abstract

Writing radiology reports is a slow and difficult task for doctors. Artificial Intelligence (AI) can help, but current models often make mistakes. They can "hallucinate" diseases that are not there, and they often sound confident even when they are wrong. Also, models trained at one hospital often fail when used at a different hospital. This research proposes a new system to fix these problems. We improve the R2Gen architecture by adding two features: (1) a "fact-checker" using Knowledge Graphs (RadGraph) to reward the model for being medically correct, and (2) an uncertainty system (Monte Carlo Dropout) that lets the model say "I am not sure." We will test this system on four large public datasets to ensure it works well in different situations.

1 Introduction

1.1 Motivation

The number of medical images taken every year is growing by 5-10%, but there are not enough radiologists to read them. This causes burnout and leads to errors in about 3-5% of cases. Current AI tools usually give a simple "Yes/No" diagnosis, which is not enough for doctors who need explanations. Vision-Language Models (VLMs) can write full reports, but they have a major flaw: they prioritize grammar over truth. They might write a sentence that looks perfect but describes a lung problem on the wrong side. Furthermore, we do not know when to trust these models because they do not provide a confidence score.

1.2 Research Objectives

This project aims to build a safer, more reliable AI for radiology. We have three main goals:

1. **Measure Uncertainty:** Use a technique called Monte Carlo Dropout to estimate how unsure the model is. This helps identify difficult cases that need a human doctor.
2. **Improve Factuality:** Change the training process so the model is punished for hallucinating medical facts, rather than just matching words.
3. **Test Robustness:** Evaluate the model on datasets from different hospitals to see if it generalizes well to new data.

2 Baselines

To measure our success, we will compare our model against these standard methods:

- **R2Gen (Chen et al., 2020) [2]:** This is a popular model for generating reports. It uses a "memory" component to remember common sentence patterns. However, it does not check if the medical facts are true.
- **CheXzero (Tiu et al., 2022) [7]:** This is the best model for "zero-shot" classification (identifying diseases without seeing them during training). It is good at detection but cannot write full reports.
- **MedCLIP (Wang et al., 2022) [8]:** This model separates image and text learning, which helps it learn from larger datasets. We use this to see how well standard contrastive learning works.

3 Datasets and Preprocessing

We will use four public datasets. We use multiple datasets to ensure the model does not just memorize one specific hospital's style.

Table 1: Summary of Selected Public Chest X-Ray Datasets

Dataset	Size (Images)	Reports?	Labels	Primary Role	Source
IU X-Ray	7,470	✓(Yes)	Manual	Training (Generation)	USA (Indiana)
MIMIC-CXR*	377,110	✓(Yes)	NLP-derived	Pre-training (Optional)	USA (Boston)
NIH ChestX-ray14	112,120	×(No)	NLP-derived	Robustness Testing	USA (NIH)
CheXpert [4]	224,316	×(No)	NLP-derived	Weakly-Sup. Training	USA (Stanford)
PadChest [1]	160,000	✓(Spanish)	Manual/NLP	Domain Shift Test	Spain

**Note: MIMIC-CXR requires credentialed access; we will prioritize IU X-Ray and CheXpert for public reproducibility unless access is granted.*

3.1 Data Sources

1. IU X-Ray (Indiana University):

- **Details:** Contains 7,470 images and 3,955 reports.

- **Importance:** This is the only fully public dataset that includes both images and text reports. We will use this for training the report generation model.
- **Views:** Includes both Frontal (PA) and Lateral views, which helps in localizing diseases.

2. NIH ChestX-ray14:

- **Details:** A large dataset with 112,120 images.
- **Importance:** It does not have reports, but it has labels for 14 diseases. We will use this to stress-test our model’s classification accuracy on new data.
- **Imbalance:** The data is imbalanced (most images are normal). We will use weighted sampling to handle this.

3. CheXpert (Stanford):

- **Details:** 224,316 images from 65,240 patients.
- **Importance:** This dataset includes an "Uncertain" label for cases where the doctor wasn’t sure. We will use this to validate our uncertainty estimation method.

4. PadChest (Spain):

- **Details:** 160,000 images with reports in Spanish.
- **Importance:** Used to test if the visual features learned by the model work across different populations and equipment types.

3.2 Preprocessing Steps

Before training, we will standardize the data:

- **Resizing:** All images will be resized to 224×224 pixels to fit the ResNet encoder.
- **Normalization:** We will normalize pixel intensity to have a mean of 0 and standard deviation of 1.
- **Text Cleaning:** All reports will be converted to lowercase. We will remove non-medical information (like patient names) and remove very rare words that appear less than 3 times.

3.3 Evaluation Metrics

We will move beyond standard NLP metrics to clinical relevance metrics:

- **RadGraph F1:** Measures if the correct clinical facts were reported.
- **AUROC:** Measures classification accuracy.
- **ECE (Expected Calibration Error):** Measures if the model’s confidence matches its actual accuracy.

4 Proposed Methodology

4.1 Method Overview

Our approach improves the standard "Encoder-Decoder" model. The Encoder turns the X-ray image into a set of numbers (vectors). The Decoder translates those numbers into a text report. We add two main improvements to this structure.

4.2 Improvement 1: Factuality Reward (Reinforcement Learning)

Standard models try to match the exact words of the training reports. This is a problem because "No pneumonia" and "Mild pneumonia" are very similar in wording but opposite in meaning.

We propose using RadGraph, a tool that extracts medical entities (like "lung", "heart") and their status (like "normal", "enlarged").

- We extract the graph from the *Ground Truth* report (G_{gt}).
- We extract the graph from the *Generated* report (G_{pred}).
- We calculate the F1-score overlap between these two graphs.
- This score is used as a Reward to train the model. If the model gets the medical facts right, it gets a high reward, even if the wording is slightly different.

4.3 Improvement 2: Uncertainty Estimation (Monte Carlo Dropout)

Deep learning models are usually deterministic (they give the same answer every time). To measure uncertainty, we use Monte Carlo (MC) Dropout.

- During training, we randomly turn off some neurons (Dropout) to prevent overfitting.
- Usually, we turn Dropout *off* during testing.
- **Our Method:** We keep Dropout *on* during testing. We run the same image through the model T times (e.g., 10 times).
- If the model generates the same report 10 times, it is Confident.
- If the model generates 10 very different reports, it is Uncertain. The variance of these outputs gives us a reliability score.

4.4 System Diagram

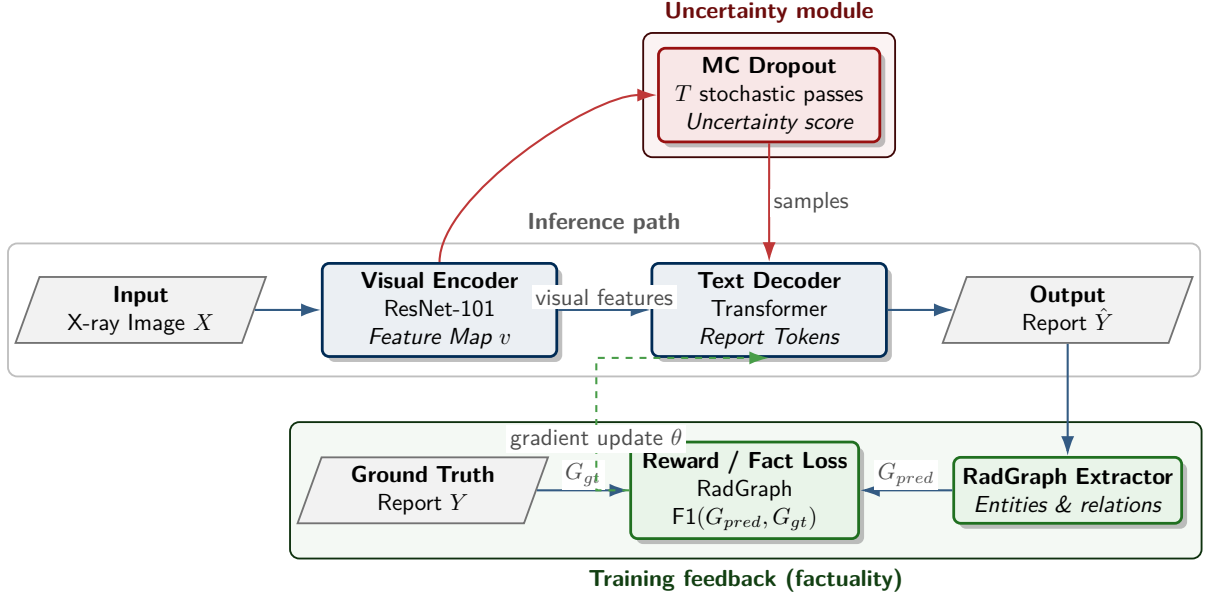


Figure 1: **System Diagram.** The inference path generates a report from an X-ray. During training, a RadGraph-based reward discourages hallucinated clinical facts. At test time, MC Dropout estimates epistemic uncertainty via stochastic forward passes.

4.5 Theoretical Formulation

The total loss function ($\mathcal{L}_{total}$) combines language fluency and clinical accuracy:

$$\mathcal{L}_{total} = \underbrace{\mathcal{L}_{CE}}_{\text{Fluency}} + \lambda \cdot \underbrace{\mathcal{L}_{Fact}}_{\text{Accuracy}} \quad (1)$$

Where:

- $\mathcal{L}_{CE}$ is the standard Cross-Entropy loss (predicting the next word).
- $\mathcal{L}_{Fact}$ is the Reinforcement Learning loss based on RadGraph F1-score.
- λ is a weight parameter. We start with $\lambda = 0$ and slowly increase it during training to stabilize learning.

Our core contribution is a hybrid loss function that balances linguistic fluency with clinical truth.

1. Epistemic Uncertainty via Bayesian Approximation: We model the neural network as a probabilistic function $p(y|x, \theta)$. Instead of a point estimate, we approximate the posterior distribution $p(\theta|D)$ using Monte Carlo (MC) Dropout. During inference, we perform T stochastic forward passes. The predictive uncertainty is quantified by the variance of the outputs:

$$\text{Var}(y) \approx \frac{1}{T} \sum_{t=1}^T (\hat{y}_t - \bar{y})^2 + \frac{1}{T} \sum_{t=1}^T \sigma_t^2 \quad (2)$$

where $\hat{y}_t$ is the prediction of the t -th pass and σ_t^2 is the inherent observation noise.

2. Factuality-Constrained Optimization: Let G_{gt} be the Knowledge Graph extracted from the ground truth report, and G_{pred} be the graph from the generated report. We define a reward function R based on the graph overlap (Precision/Recall of clinical entities). The final loss function $\mathcal{L}_{total}$ combines Cross-Entropy (CE) loss with a Policy Gradient (RL) loss:

$$\mathcal{L}_{total} = \underbrace{-\sum \log p(y_t|y_{<t}, x)}_{\text{Linguistic Fluency}} + \underbrace{\lambda \mathbb{E}_{\hat{y} \sim p_\theta} [1 - \text{F1}_{\text{RadGraph}}(G_{pred}, G_{gt})]}_{\text{Clinical Factuality}} \quad (3)$$

where λ is a hyperparameter balancing fluency and factuality.

4.6 Implementation Steps

Algorithm 1: Training and Testing Logic

Input: Images $\mathcal{X}$, Real Reports $\mathcal{Y}$

Output: Trained Model θ

- 1 **1. Setup:** Load Pre-trained ResNet and BERT tokenizer
 - 2 **while** *Training is not done* **do**
 - 3 **2. Forward Pass:**
 - 4 Get image features $v = \text{Encoder}(x)$
 - 5 Generate text report $\hat{y} = \text{Decoder}(v)$
 - 6 **3. Calculate Language Loss:**
 - 7 Compare words of $\hat{y}$ and y (Cross Entropy Loss)
 - 8 **4. Calculate Fact Loss (RL):**
 - 9 Extract facts: $G_{pred} = \text{RadGraph}(\hat{y})$
 - 10 Extract facts: $G_{real} = \text{RadGraph}(y)$
 - 11 Calculate Reward = F1 Score between G_{pred} and G_{real}
 - 12 **5. Update Model:**
 - 13 Adjust weights θ to maximize Reward and minimize Language Loss
 - 14 **end**
 - 15 **6. Testing (Uncertainty):**
 - 16 Run image T times with Dropout ON
 - 17 Calculate variance of outputs to measure confidence
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5 Project Timeline

- **Week 1-2:** Setup environment and reproduce CheXzero baseline on IU X-Ray.
- **Week 3:** Implement Uncertainty Quantification (MC Dropout).
- **Week 3-4:** Implement RadGraph-based loss function (Algorithm 1).
- **Week 4-5:** Run cross-dataset validation on NIH ChestX-ray14.

- **Week 5:** Analyze failure modes and write final paper.

6 Conclusion

By adding a "fact-checker" and an "uncertainty-checker," we address the biggest problems in medical AI today. We move beyond simple accuracy scores to create a model that is robust, truthful, and knows when to ask a human for help. This is a crucial step towards safely using AI in real hospitals.

References

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